



Sampling Distributions I

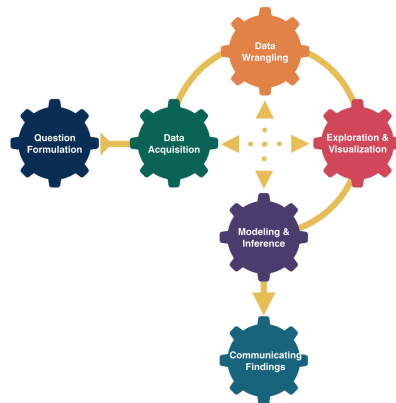
Megan Ayers
 Stat 141 | Fall 2026
 Wednesday, Week 6

Reminder

- Quiz 4 tomorrow during lab, but you can leave right after (no lab assignment)

Goals for Today

- Start learning the foundations of inference
- Perform a group sampling activity
- Discuss random sampling: the heart of statistics!



Distinguishing between the population and the sample

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p + \epsilon$$

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \dots + \hat{\beta}_p x_p$$

Parameters:

- Based on the **population**
- Unknown then if don't have data on the whole population
- EX: $\beta_0, \beta_1, \dots, \beta_p$
- EX: μ (population mean) p (population proportion)

Statistics:

- Based on the **sample** data
- Known
- Usually estimate a population parameter
- EX: $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p$
- EX: $\bar{x}$ (sample mean), $\hat{p}$ (sample proportion)

General Definitions: Parameters and Statistics

- **Parameter:** Numerical characteristic of a population (e.g., average of a variable in a population)
- **Statistic:** Estimate of the population parameter using the sample (e.g., average of the same variable *in the sample*)
- Researchers often wish to investigate the value of a **parameter** in a population.
 - The proportion of U.S. voters who plan to vote for a particular presidential candidate.
 - The mean life-time earnings for Reed college graduates
- But it is often not feasible to collect complete information on the population.
- Instead, researchers collect a sample and measure a **statistic**, which *estimates* the population parameter
 - The proportion of voters in a sample of size 500 who plan to vote for the candidate.
 - The mean life-time earnings for 100 randomly chosen Reed graduates.

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Sampling Activity

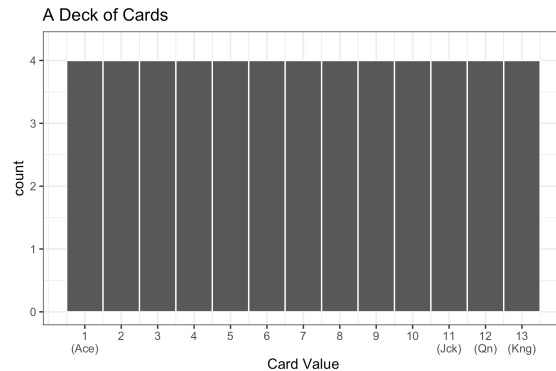
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Decks of cards

If we count:

- **Aces** as 1
- **Jacks** as 11
- **Queens** as 12
- **Kings** as 13

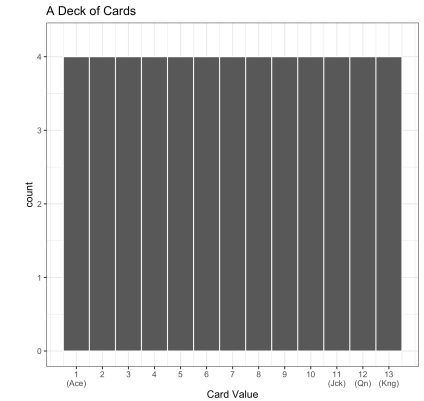
The distribution of numbers in a deck of cards looks like:



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Drawing cards: Population, Sample, Parameter, and Statistic

- Today, we're going to be:
 - Randomly drawing 10 cards from a deck of cards
 - Calculating the average value of our 10 cards
- **Q:** What is the **sample**? What is the **population**?
- **Q:** What is the **statistic**? What is the **parameter**?



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Activity Instructions

1. Thoroughly shuffle your group's deck of cards.
2. Draw 10 cards from the deck **without replacement** to form a sample.
3. Compute the average/mean value of your 10 cards
4. Add the value of the average/mean as a dot on the chalkboard.
5. Repeat steps 1 - 4 an additional **seven** times.

Each group should have calculated 8 averages from 8 different samples of 10 cards!

Small Group Discussion

Once you're done, discuss the following questions with your group:

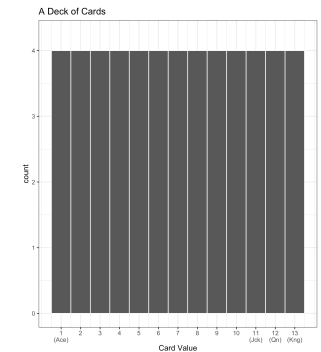
1. What is the true average card value in a deck of cards?
2. How does the distribution of sample means compare to the distribution of card values in a deck of cards?
3. What is the relationship between the centers of the two distributions?
4. Which distribution appears to have more variability?
5. How do the shapes of the two distributions compare? Why do they differ?

15:00

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Discussion

1. What is the true average card value in a deck of cards?
2. How does the distribution of sample means compare to the distribution of card values in a deck of cards?
3. What is the relationship between the centers of the two distributions?
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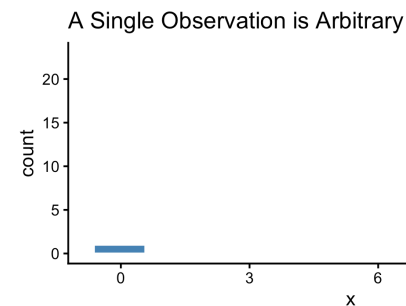


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Sampling Overview

Sampling Overview

- The distribution of a data set allows us to quantify the shape, center, and spread of the data.
- While a single observation in a data set may appear arbitrary...

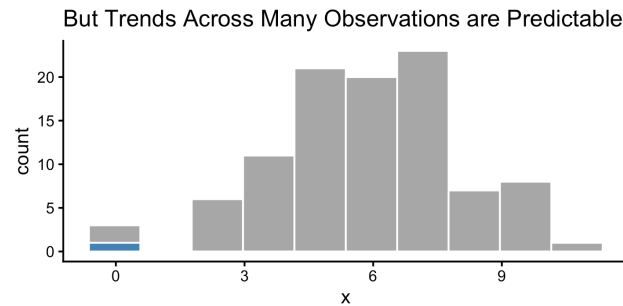


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Sampling Overview

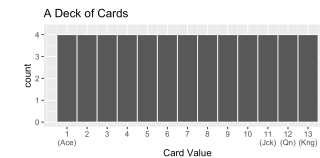
- The distribution of a data set allows us to quantify the shape, center, and spread of the data.
- While a single observation in a data set may appear arbitrary... repeated trials often show that outcomes follow certain patterns.



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Sampling Overview

- We know that a variable from a population has a distribution
 - e.g., the distribution of card values in a deck of cards

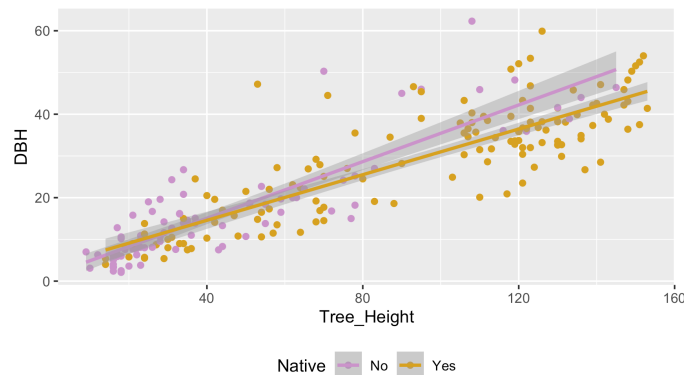


- Moral of Today:** Statistics (e.g., mean of variable in a sample) have distributions too!!
 - How? We only have ONE statistic – the statistic from the ONE sample we drew
 - Distribution is **over all the possible samples we could have drawn** (we just see 1)
- Implication:** Statistics themselves have a *mean*, *standard deviation*, *5-number summary*
 - The mean tells us the statistic's typical value in a randomly chosen sample.
 - The standard deviation tells us how the statistic fluctuates from sample to sample.
- Very Powerful:** e.g., Can use this distribution to give **plausible** ranges for the parameter and a sense of **uncertainty** in a given statistic.

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Bigger Picture - Quantifying Our Uncertainty

R has been giving us uncertainty estimates (ex. `geom_smooth` when we don't set `se = FALSE`):



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Bigger Picture - Quantifying Our Uncertainty

R has been giving us uncertainty estimates (ex. `std_error` in summaries from `lm()`):

```
# A tibble: 4 × 7
  term      estimate std_error statistic p_value lower_ci upper_ci
<chr>      <dbl>      <dbl>      <dbl>   <dbl>   <dbl>   <dbl>
1 intercept  7.39         3.65        2.03  0.044    0.201   14.6
2 DBH        2.25         0.17       13.3    0       1.92    2.59
3 Native: Yes 11.1         5.59        1.98  0.049    0.067   22.1
4 DBH:NativeYes 0.315       0.215        1.47  0.144   -0.108    0.739
```

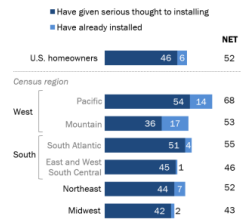
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Bigger Picture - Quantifying Our Uncertainty in Statistics

Uncertainty estimates are constantly reported in news and journal articles:

More than four-in-ten U.S. homeowners are considering residential solar panels

% of U.S. homeowners who say they ____ solar panels at home



Note: Based on homeowners. Respondents who gave other responses or did not give an answer are not shown.
Source: Survey conducted Oct. 1-13, 2019.

PEW RESEARCH CENTER

Note: The findings are based on a [survey](#) conducted Oct. 1-13, 2019, among 3,627 U.S. adults on Pew Research Center's American Trends Panel. The margin of sampling error for the full sample is plus or minus 2.1 percentage points. The margin of error for the 2,564 U.S. homeowners is plus or minus 2.5 percentage points. See [full timeline results](#).

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Bigger Picture - Quantifying Our Uncertainty in Statistics

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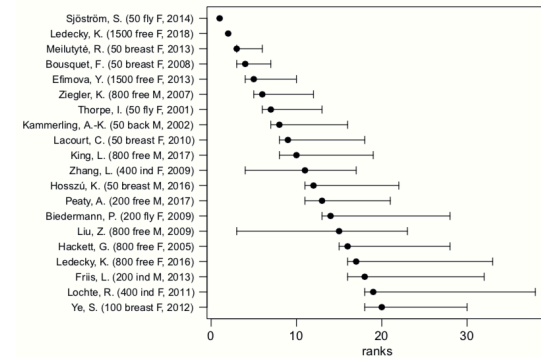


Figure 2: Ranking of the top 20 personal best swims (and their swimmers), 2001-2018, with 95% confidence intervals. Better-ranked swimmers are higher on the y-axis.

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Recap: Sampling Distributions and Statistical Inference

- **Goal:** Draw conclusions about the population based on the sample.
 - We've seen how to calculate a statistic from our sample
 - But different samples give different statistics: how do we know whether to trust the one we have?
- **Sampling distributions** show how widely our statistic can range across samples
 - This helps us understand how much to trust any single statistic
 - We'll spend a couple more days learning about sampling distributions, then will learn how to use them to make inferences!

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